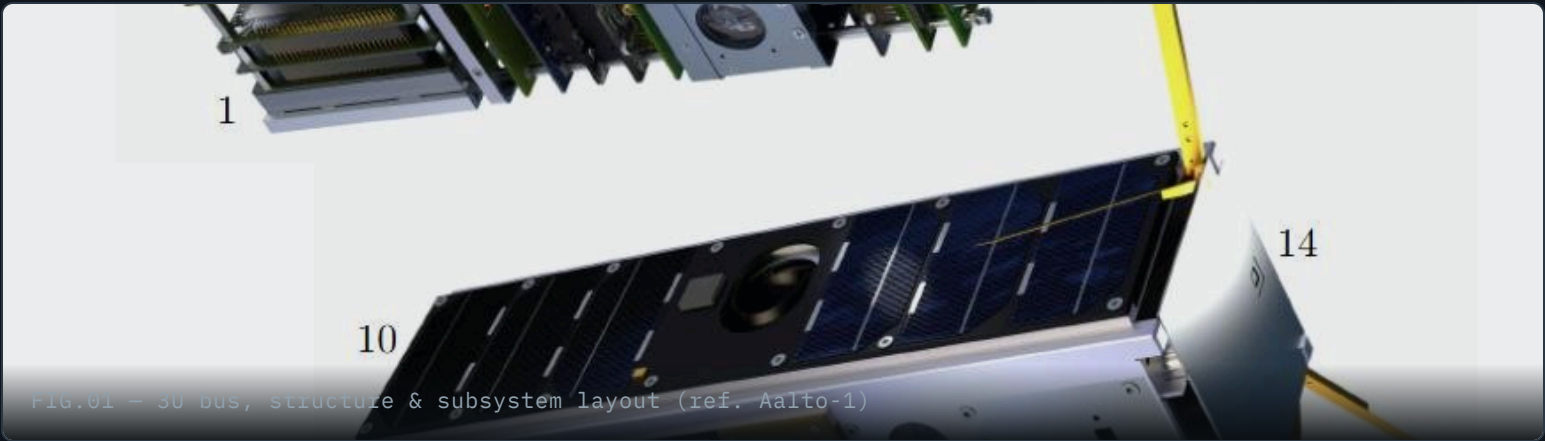


CubeSat Reliability Analysis

A full Failure Mode & Effects Analysis on a 3U, Aalto-1-class CubeSat — mapping 30 failure modes across 9 subsystems, then engineering the corrective actions that cut total system risk by more than half.



30

FAILURE MODES ASSESSED

9

SUBSYSTEMS & PAYLOADS

-55%

REDUCTION IN TOTAL RPN

280 → 105

PEAK RPN, BEFORE → AFTER

ROLE

Team of 3. My scope: Onboard Computer, Electrical Power, and Structure/Thermal subsystem analysis — co-developed the corrective-action research and Pareto prioritization with teammates Pragya Mallick and Pegah Haghshomar.

CONTEXT

ELEC-E8712, Design for Reliability — Aalto University, Assignment 1, Oct 2024. Modeled on Aalto-1, Finland's first satellite.

TOOLS

- FMEA
- Pareto Analysis
- RPN Scoring
- Excel

This project applies **Failure Mode & Effects Analysis (FMEA)** to a 3U CubeSat modeled on **Aalto-1** — Finland's first satellite, built by Aalto University and launched in 2017. Using a real, publicly documented spacecraft as the subject let us practice reliability engineering against genuine hardware constraints rather than a hypothetical system.

The spacecraft carries three payloads — a hyperspectral **Spectral Imager**, a **Radiation Monitor (RADMON)**, and an experimental **Electrostatic Plasma Brake (EPB)** for deorbiting — on top of the platform subsystems every CubeSat needs: an onboard computer, electrical power, attitude determination & control, communications, thermal control, and the structural bus that holds it all together.

My role covered the **Onboard Computer, Electrical Power, and Structure/Thermal** analysis, and I co-developed the corrective-action research and the Pareto-based prioritization with my teammates, Pragya Mallick and Pegah Haghshomar.

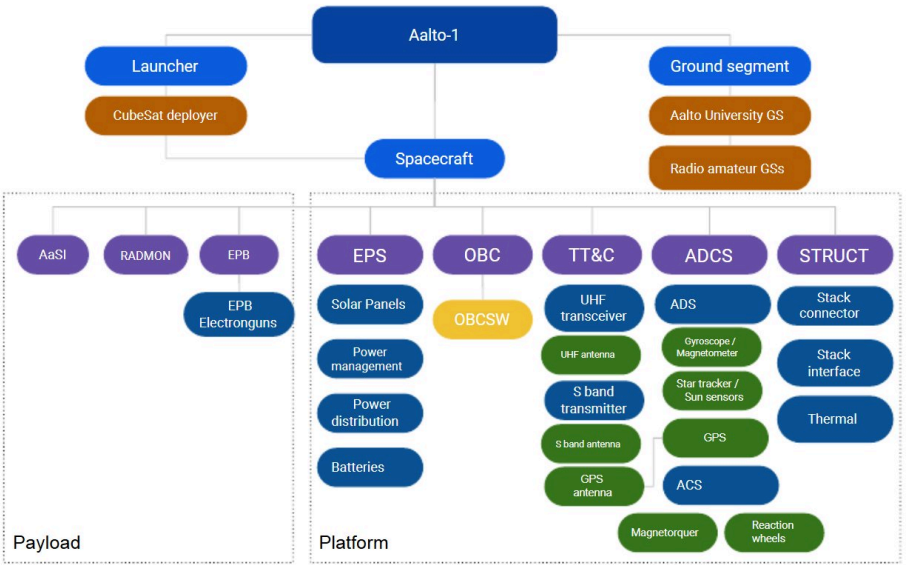
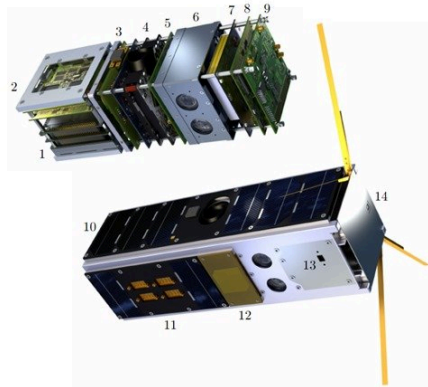


FIG.02 – Product tree: payload vs. platform subsystem breakdown



Spectral Imager (AaSI) – hyperspectral Earth-observation payload



Deployable solar panel – primary EPS power source

02 METHODOLOGY

How the analysis works

For every subsystem function, we identified plausible failure modes, their effects, and root causes, then scored each on a 1–9 scale before ranking and prioritizing with a Pareto analysis.

$$\begin{array}{ccccc} \text{Severity} & \times & \text{Occurrence} & \times & \text{Detection} & = & \text{RPN} \\ \text{IMPACT IF IT HAPPENS} & & \text{LIKELIHOOD} & & \text{ODDS IT'S CAUGHT IN TIME} & & \end{array}$$

01

Identify

Break each subsystem into functions and list plausible failure modes and mechanisms for each.

02

Score

Rate severity, occurrence and detection 1–9 against current controls; multiply for a Risk Priority Number.

03

Prioritize

Rank all failure modes by RPN and run a Pareto analysis to find the small set driving most of the risk.

04

Mitigate & re-score

Design a corrective action for each priority item, then re-score to verify the RPN actually dropped.

03 RISK ASSESSMENT – BEFORE CORRECTIVE ACTION

30 failure modes, ranked

The initial sweep across all subsystems surfaced 30 distinct failure modes. Sorted by RPN, the distribution is classically Pareto-shaped: a handful of failure modes — connector mounting, power regulation, and bus communications among them — account for a disproportionate share of total system risk.

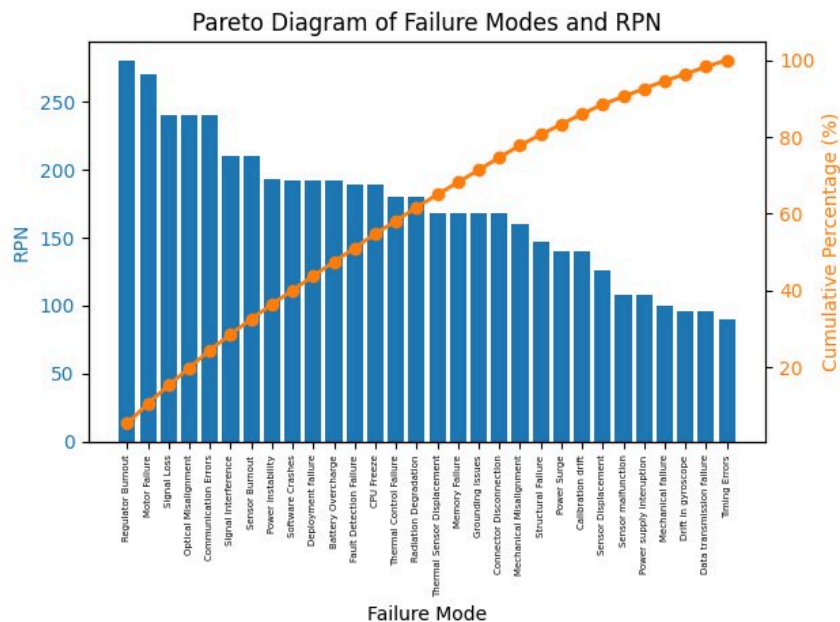


FIG.03 – Pareto ranking of all 30 failure modes by RPN, with cumulative-risk curve

Highest-risk failure modes

Top 8 by RPN, scale relative to the highest score recorded (280).

Power Distribution Unit – connectors

Connector disconnection under launch vibration



S8 · 07 · D5

280

PMU (Power Management Unit)

Battery overcharging / power regulation failure



S9 · 06 · D5

270

Data Communication Bus (RS-485)

Bus overload causing comms & data loss



S8 · 05 · D6

240

Fabry-Pérot Interferometer

Optical misalignment from thermal/mechanical stress



S8 · 06 · D5

240

Solar Panels

Radiation degradation & thermal cracking



S8 · 06 · D5

240

GPS Antenna

Signal interference from poor placement/shielding



S7 · 06 · D5

210

Sun Sensors — thermal burnout

Sensor burnout from prolonged sun exposure



S7 · 06 · D5

210

ONBOARD COMPUTER

Power Supply to the OBC

Voltage fluctuation / insufficient power delivery



S8 · 04 · D6

192

RPN before corrective action

04 CORRECTIVE ACTIONS

Five failure modes, fully researched

Every failure mode in the FMEA got a recommended action. For five of the highest-priority ones we went further, researching and justifying a specific engineering fix with a cost-benefit case and supporting evidence — the level of depth a real design review would expect.

Voltage regulator burnout

192 → 60

ACTION

Dual-layer voltage regulation: linear regulators paired with transient-voltage-suppressor (TVS) diodes.

WHY IT WORKS

Combines steady regulated output with clamping protection against spikes; if one regulation path fails, the other keeps the bus alive instead of the whole system going dark.

EVIDENCE CITED

TVS protection reported to cut regulator burnout incidence by up to 65% in comparable systems.

Connector disconnection (launch vibration)

280 → 80

ACTION

Switch from adhesive-mounted connectors to locking connectors with integrated strain relief, plus added vibration qualification testing.

WHY IT WORKS

Adhesive mounting wasn't rated for launch-level vibration loads. A mechanical lock and strain relief remove the failure path rather than just dampening it — soldered alternatives were ruled out on weight and serviceability.

EVIDENCE CITED

Locking connectors with strain relief reported to cut connector failure rates by up to 80% in high-vibration environments.

Grounding issues / short circuits

168 → 50

ACTION

Redesigned grounding architecture around a star topology, using corrosion-resistant (gold-plated) grounding hardware.

WHY IT WORKS

Star grounding removes the shared return paths that cause ground loops and cross-talk; corrosion-resistant hardware keeps that connection reliable through years of thermal cycling in vacuum.

EVIDENCE CITED

Star-grounding configurations reported to reduce short-circuit incidence by up to 70% in CubeSat-class systems.

Structural overload / cracking

147 → 60

ACTION

Replace select aluminum structural members with carbon-fiber-reinforced polymer (CFRP).

WHY IT WORKS

CFRP's strength-to-weight ratio lets the structure absorb higher loads without an equivalent mass penalty — a real trade on a mass-constrained 3U bus, offset against higher material cost and machining sensitivity.

EVIDENCE CITED

CFRP components reported to tolerate ~30% higher loads than aluminum at equal weight.

Sun sensor burnout (prolonged sunlight exposure)

210 → 64

ACTION

UV-resistant coatings plus phase-change-material (PCM) thermal buffering around the sensor electronics.

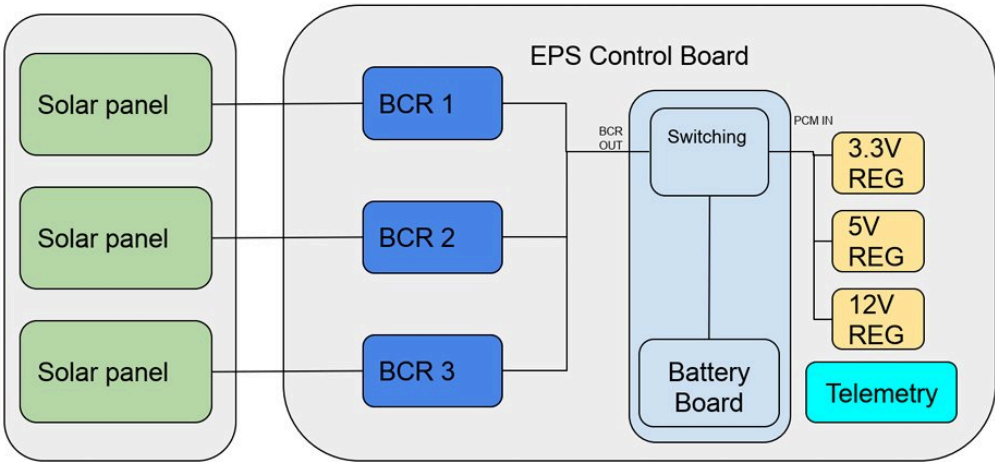
WHY IT WORKS

The coating slows optical and electronic degradation from constant solar exposure, while the PCM absorbs peak heat loads instead of letting them reach the sensor.

EVIDENCE CITED

UV-resistant coatings reported to cut sensor degradation rate by up to 50% under continuous sun exposure.

RPN values reflect the FMEA worksheet's before/after scoring for each corresponding failure mode. Supporting figures as cited in the team's corrective-actions report.



EPS block diagram – where the regulator & connector fixes apply



Bus structure cutaway – grounding path & structural members

After incorporating the corrective actions, every failure mode was re-scored under the same Severity × Occurrence × Detection method. Total system risk — the sum of every RPN in the worksheet — fell by more than half, and every failure mode that had exceeded the 200-RPN action threshold dropped below it.

5,179 → 2,331

Total RPN across all 30 failure modes (–55%)

172.6 → 77.7

Average RPN per failure mode

7 → 0

Failure modes above the 200-RPN action threshold

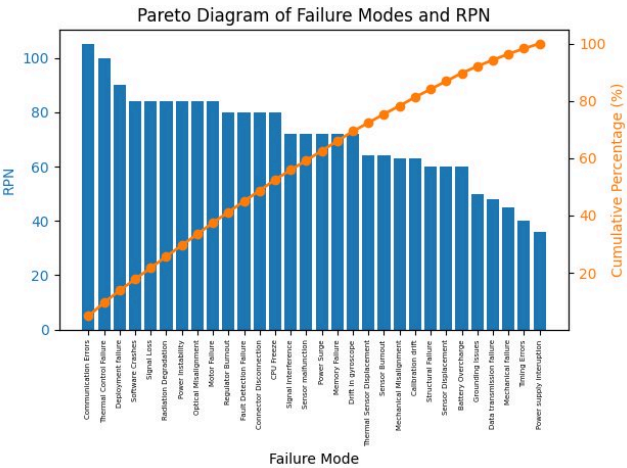


FIG.04 – Pareto ranking after corrective action

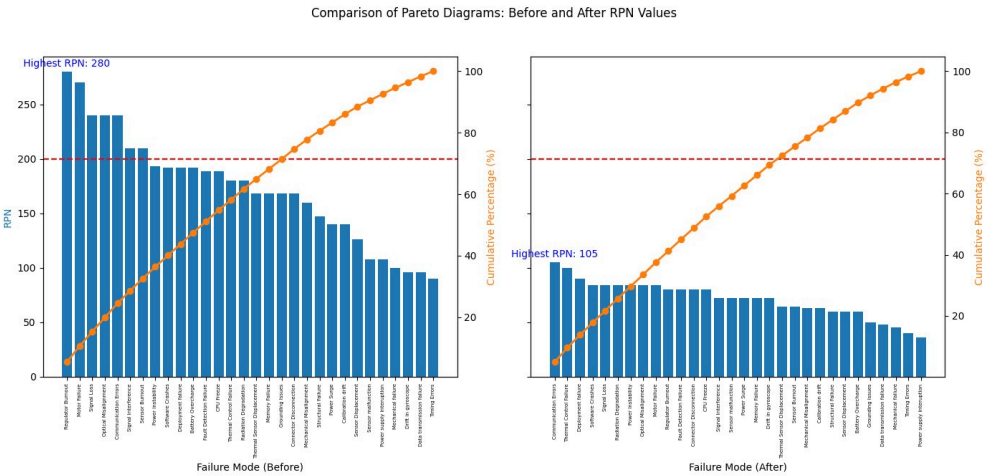
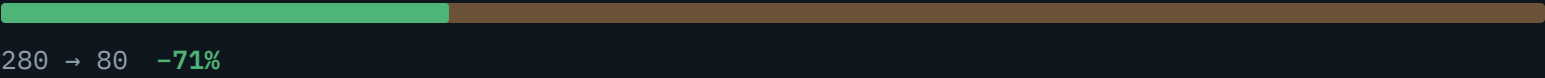


FIG.05 – Before / after comparison, highest RPN 280 → 105

Same 8 failure modes, before vs. after

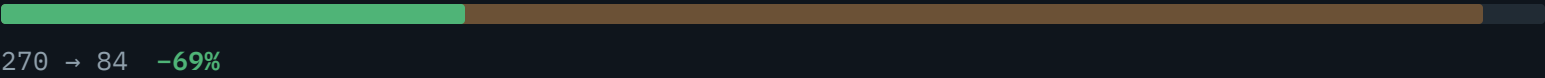
ELECTRICAL POWER

Power Distribution Unit – connectors



ELECTRICAL POWER

PMU (Power Management Unit)



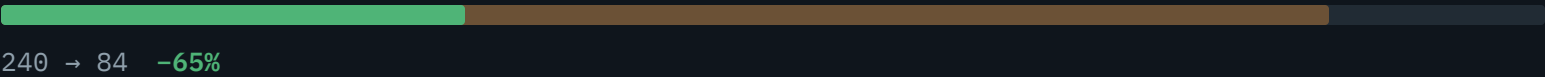
ONBOARD COMPUTER

Data Communication Bus (RS-485)



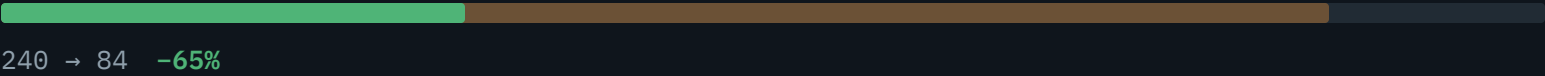
PAYLOAD – SPECTRAL IMAGER

Fabry-Pérot Interferometer



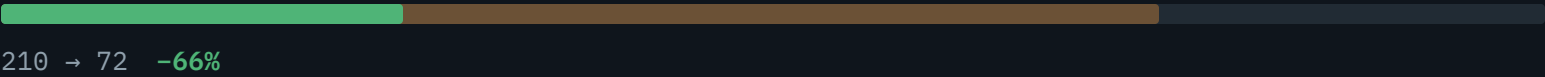
ELECTRICAL POWER

Solar Panels



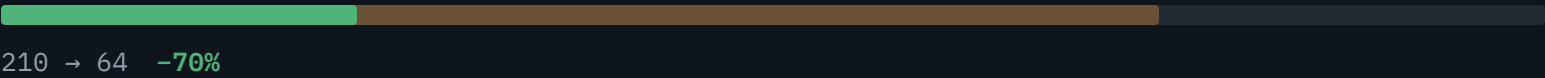
COMMUNICATIONS

GPS Antenna



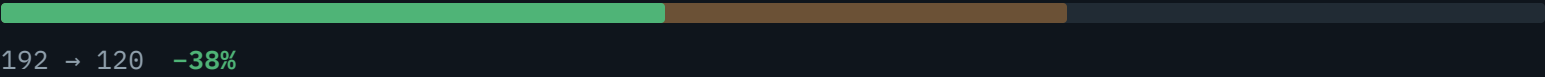
ADCS

Sun Sensors – thermal burnout



ONBOARD COMPUTER

Power Supply to the OBC



Before After

View full FMEA worksheet – all 30 failure modes

Fixing the top failure mode without adding mass or cost

The highest-RPN item in the whole worksheet — Power Distribution Unit connector disconnection under launch vibration, RPN 280 — needed a fix that survived a 3U CubeSat's tight mass and cost budget, not just the highest theoretical reliability gain.

CONSIDERED

Adhesive-mounted connectors (the original design, not rated for launch-level vibration loads) and soldered joints (more vibration-resistant, but heavier and harder to service or rework post-integration).

CHOSEN APPROACH

Locking connectors with integrated strain relief, plus added vibration qualification testing.

WHY

A mechanical lock and strain relief remove the failure path entirely rather than just damping vibration — reported to cut connector failure rates by up to 80% in high-vibration environments. Soldered alternatives were ruled out specifically on weight and serviceability, which matter as much as raw reliability on a mass- and integration-constrained CubeSat bus.

What made this analysis genuinely hard

Scoring risk without proprietary flight data

Aalto-1 is a real, publicly documented spacecraft, but detailed component-level failure statistics from its actual flight aren't public. Severity, occurrence, and detection scores had to be grounded in comparable-system literature and engineering judgment rather than the satellite's own telemetry — a limitation worth stating explicitly rather than letting the RPN numbers imply more precision than they have.

Dividing scope across three people without gaps or double-counting

My subsystems — Onboard Computer, Electrical Power, and Structure/Thermal — interact constantly (power delivery to the OBC, thermal control of both). Keeping failure-mode ownership clean at those boundaries, and agreeing with teammates on where one person's analysis ended and another's began, took as much coordination as the technical scoring itself.

08

REFLECTION

What I'd do with another pass at this

Next steps

The five researched corrective actions were each evaluated as if independent, but some plausibly interact — the star-grounding redesign for short-circuit risk touches the same wiring harness as the connector fix, for instance. A natural next step would be building lightweight fault trees for the highest-RPN items to check whether stacking these mitigations changes the combined risk picture, rather than assuming the individual RPN reductions simply add up.

09

SUMMARY

What this project demonstrates

- Systematic failure identification across mixed hardware, software and mechanical subsystems
- Evidence-based corrective-action design, including cost-benefit trade-offs
- Reading and reasoning about real spacecraft subsystem documentation
- Quantitative risk prioritization using RPN scoring and Pareto analysis
- Closed-loop verification — re-scoring every fix to confirm the risk actually dropped
- Clear technical communication of risk data to a non-specialist audience